

Review

Digital Twin Technology for Urban Flood Risk Management: A Systematic Review of Remote Sensing Applications and Early Warning Systems

Mohammed Hlal ^{1,*}, Jean-Claude Baraka Munyaka ², Jérôme Chenal ^{1,2}, Rida Azmi ¹, El Bachir Diop ¹, Mariem Bounabi ¹, Seyid Abdellahi Ebnou Abdem ¹, Mohamed Adou Sidi Almouctar ¹ and Meriem Adraoui ³

¹ Center of Urban Systems—CUS, Mohammed VI Polytechnic University—UM6P, Benguerir 43150, Morocco; jerome.chenal@epfl.ch (J.C.); rida.azmi@um6p.ma (R.A.); elbachir.diop@um6p.ma (E.B.D.); mariem.bounabi@um6p.ma (M.B.); seyd.ebnouabdem@um6p.ma (S.A.E.A.); almouctar.mohamed-adousidi-ext@um6p.ma (M.A.S.A.)

² Urban and Regional Planning Community (CEAT), Ecole Polytechnique Fédérale de Lausanne (EPFL), 1015 Lausanne, Switzerland; baraka.munyaka@epfl.ch

³ Faculty of Arts and Humanities, Moulay Ismail University, Meknes 50050, Morocco; m.adraoui@umi.ac.ma

* Correspondence: mohammed.hlal@um6p.ma

Abstract

Digital Twin (DT) technology has emerged as a transformative tool in urban flood risk management (UFRM), enabling real-time data integration, predictive modeling, and decision support. This systematic review synthesizes existing literature to evaluate the scientific impact, technological advancements, and practical applications of DTs in UFRM. Using the PRISMA 2020 framework, we retrieved 1085 records (Scopus = 85; Web of Science = 1000), merged and deduplicated them using DOI and fuzzy-matched titles, screened titles/abstracts, and assessed full texts. This process yielded 85 unique peer-reviewed studies published between 2018 and 2025. Key findings highlight the role of remote sensing (e.g., satellite imagery, IoT sensors) in enhancing DT accuracy, the integration of machine learning for predictive analytics, and case studies demonstrating reduced flood response times by up to 40%. Challenges such as data interoperability and computational demands are discussed, alongside future directions for scalable, AI-driven DT frameworks. This review identifies key technical and governance challenges while recommending the development of modular, AI-driven DT frameworks, particularly tailored for resource-constrained regions.

Keywords: digital twin; urban flood risk management; remote sensing; machine learning; deep learning; climate resilience; smart cities; early warning systems



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1. Introduction

Flooding is one of the most destructive and recurrent natural hazards globally, resulting in widespread socioeconomic disruptions, critical infrastructure failures, displacement, and loss of life [1]. According to global disaster databases, between 1993 and 2024, flood events have affected more than 2.3 billion people [2]. Also, flood events resulted in over USD 400 billion in economic damages from 2010 to 2024 [3]. Projections suggest that by 2050, the escalating frequency and intensity of climate-induced disasters could impose economic losses of up to 14.5 million deaths and USD 12.5 trillion in economic losses worldwide, making flood resilience a critical concern for national governments, urban planners, and global institutions [4].

Urban areas are particularly vulnerable due to their dense population, impervious surface coverage, and ageing or insufficient drainage infrastructure [5]. The phenomenon of “pluvial flooding” caused by intense rainfall overwhelming stormwater systems has become increasingly common in megacities like Chiang Mai in Thailand and Lagos in Nigeria [6,7].

Anthropogenic factors further exacerbate this vulnerability. Unregulated urban sprawl, deforestation, and the encroachment on natural floodplains reduce the absorptive capacity of the urban environment [8]. Moreover, disparities in infrastructure investment and institutional capacity, particularly in low- and middle-income countries lead to disproportionate exposure among marginalized communities [9]. These spatial and socioeconomic inequalities make urban flood risk not just an environmental issue, but also a question of climate justice [10,11].

Despite the rising risks, traditional urban flood risk management (UFRM) strategies remain largely reactive, fragmented, and technologically outdated [12]. Conventional hydrological models are static and reliant on historical data, offering limited utility in dynamic urban environments where flood risk is continuously evolving [13]. Similarly, top-down policy frameworks often fail to capture local realities or integrate multi-source, real-time data [14]. As a result, early warning systems and emergency response mechanisms are frequently delayed, ineffective, or unequally distributed for undermining efforts to build urban resilience.

Amid this context, Digital Twin (DT) technology has emerged as a promising tool to revolutionize UFRM. A Digital Twin is a real-time, dynamic digital replica of a physical system, which continuously updates using real-world data streams from sensors, satellites, and other inputs [15]. Originating from the manufacturing sector, DTs are increasingly applied to critical infrastructure, smart cities, and climate adaptation systems due to their ability to model complex, multi-scale interactions [16].

In flood risk applications, DTs enable:

- Predictive simulations under multiple hazard scenarios (e.g., simultaneous fluvial and pluvial flooding, storm surge combined with river overflow)
- Integration of heterogeneous data sources, including satellite remote sensing, in situ sensors, social media, and crowdsourced inputs. Social media refers to platforms like Twitter or Facebook used for rapid situational updates, while crowdsourced inputs may include structured community reports via dedicated apps or local participatory mapping platforms.
- AI-driven analytics for forecasting and anomaly detection
- Interactive dashboards and 3D environments to support decision-making

The convergence of remote sensing (RS), Internet of Things (IoT), Artificial Intelligence (AI), and cloud computing has made DTs increasingly viable for real-time, spatially explicit flood monitoring [17]. By synthesizing massive volumes of data into actionable intelligence, DTs support a proactive and anticipatory model of disaster management far superior to reactive post-event interventions [18].

While Digital Twin (DT) frameworks are increasingly applied across hydrological domains ranging from urban stormwater systems to large-scale river basins, there remains a conspicuous lack of unified architectural standards, integrated technological taxonomies (e.g., IoT, AI, GIS), and operational interoperability [19]. Notable implementations such as Bentley’s Open Flows Flood and the European Commission’s Destination Earth illustrate the methodological heterogeneity and divergent scope of existing systems [20]. This fragmentation underscores the urgency for standardized, modular frameworks capable of guiding scalable and context-aware DT deployments [21].

Current DT applications in urban flood risk management (DT-UFRM) predominantly emphasize technical capabilities such as real-time sensor integration, hydraulic simulation, and machine learning-driven forecasting, while often overlooking crucial dimensions of policy coherence, climate justice, and equitable risk communication [22]. Case studies like Porto's H2Porto and China's Digital Twin River Basin initiative exemplify infrastructure-centric designs that marginalize social inclusivity, particularly in vulnerable or underserved communities [23,24]. This gap risks reinforcing digital inequities, particularly in the Global South (Developing countries), where access to DT-generated early warnings and participatory governance mechanisms remains limited.

Moreover, existing systematic reviews and empirical studies are disproportionately biased toward high-income, data-rich contexts such as European smart cities and U.S. federal flood forecasting systems neglecting the needs of flood-prone but data-constrained regions [25]. Examples such as the sediment-laden Yellow River Basin or Kerala's monsoon-vulnerable infrastructure demand localized, resource-sensitive DT adaptations that remain underrepresented in scholarly discourse [26].

Despite the transformative potential of remote sensing (RS) technologies ranging from satellite-based heat island mapping to UAV-enabled 3D city modeling their integration into DT-UFRM architectures is often fragmented and unsystematic [27]. Initiatives like SURE (Satellite-based Urban Resilience Enhancement) demonstrate the promise of multi-platform RS fusion for enhancing simulation fidelity [28]. Yet, the broader academic and operational discourse lacks a critical, comparative analysis of scalable, low-cost RS solutions that can function under real-time hazard conditions, especially in low-resource settings [29].

Escalating climate-induced disasters (e.g., Pakistan's 2022 megafloods) necessitate DT frameworks that bridge hydrological science, urban planning, and social governance [30]. While strategies such as the U.S. National Digital Twin R&D Strategic Plan advocate for cross-sectoral integration, the prevailing DT-UFRM landscape remains fragmented and siloed limiting the potential of these systems to support holistic, anticipatory resilience planning [31].

To address these knowledge gaps, this systematic review aims to critically assess the state of the art, practical deployment, and theoretical evolution of Digital Twin technology in the context of global urban flood risk management. This systematic review synthesizes insights from a curated set of 85 peer-reviewed studies, selected from an initial pool of 1085 articles retrieved from Scopus and Web of Science between 2018 and 2025. These studies span disciplines such as hydrology, geoinformatics, disaster risk science, and public policy.

The specific objectives are to:

- Analyze the technical architectures of DTs used in flood modeling, with an emphasis on RS-IoT-AI integration.
- Evaluate real-world case studies from diverse geopolitical and hydrological contexts.
- Identify the challenges and limitations of technological, institutional, and ethical hindering DT deployment.
- Explore the policy and governance frameworks surrounding DT adoption, including transboundary cooperation and climate equity.
- Recommend future research directions, including AI-driven innovations, quantum computing, and participatory modeling.

2. Materials and Methods

2.1. Systematic Review Framework

This study adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines, which provide a structured approach for ensuring transparency, replicability, and comprehensiveness in systematic reviews [32]. The review

design follows a four-step protocol: identification, screening, eligibility assessment, and inclusion, tailored to the multidisciplinary nature of Digital Twin (DT) applications in urban flood risk management (UFRM).

The methodology was intentionally designed to capture a diverse, globally representative, and technically rigorous sample of peer-reviewed research spanning multiple scientific domains, including hydrology, geospatial science, computer vision, climate adaptation, and public policy.

2.2. Data Sources and Search Strategy

To construct a comprehensive and methodologically robust evidence base, the literature was systematically retrieved from two major academic databases: Scopus and Web of Science. This search yielded an initial pool of 1085 documents, comprising 85 records from Scopus and 1000 from Web of Science. The choice of these databases was informed by their broad disciplinary coverage, particularly in engineering, environmental sciences, remote sensing technologies, and climate-related interdisciplinary research. Their stringent peer-review standards and extensive indexing practices assured the reliability and scholarly quality of the materials collected. This discrepancy is due to differences in indexing scope; Web of Science includes a wider range of environmental science and engineering journals, while Scopus yielded fewer flood-focused DT publications for the selected timeframe.

The review focused on studies published between 2018 and 2025, a timeframe marked by accelerated development in Digital Twin (DT) technologies and heightened global attention to climate resilience and urban flood management strategies. This period was selected to capture both foundational advances and the most recent trends in DT applications.

To uphold scientific rigor, the review strictly excluded gray literature, preprints, unpublished technical reports, and other non-peer-reviewed sources. Only articles published in reputable, peer-reviewed journals with transparent methodological disclosures were included, ensuring the reliability and replicability of the synthesized findings (Figure 1).

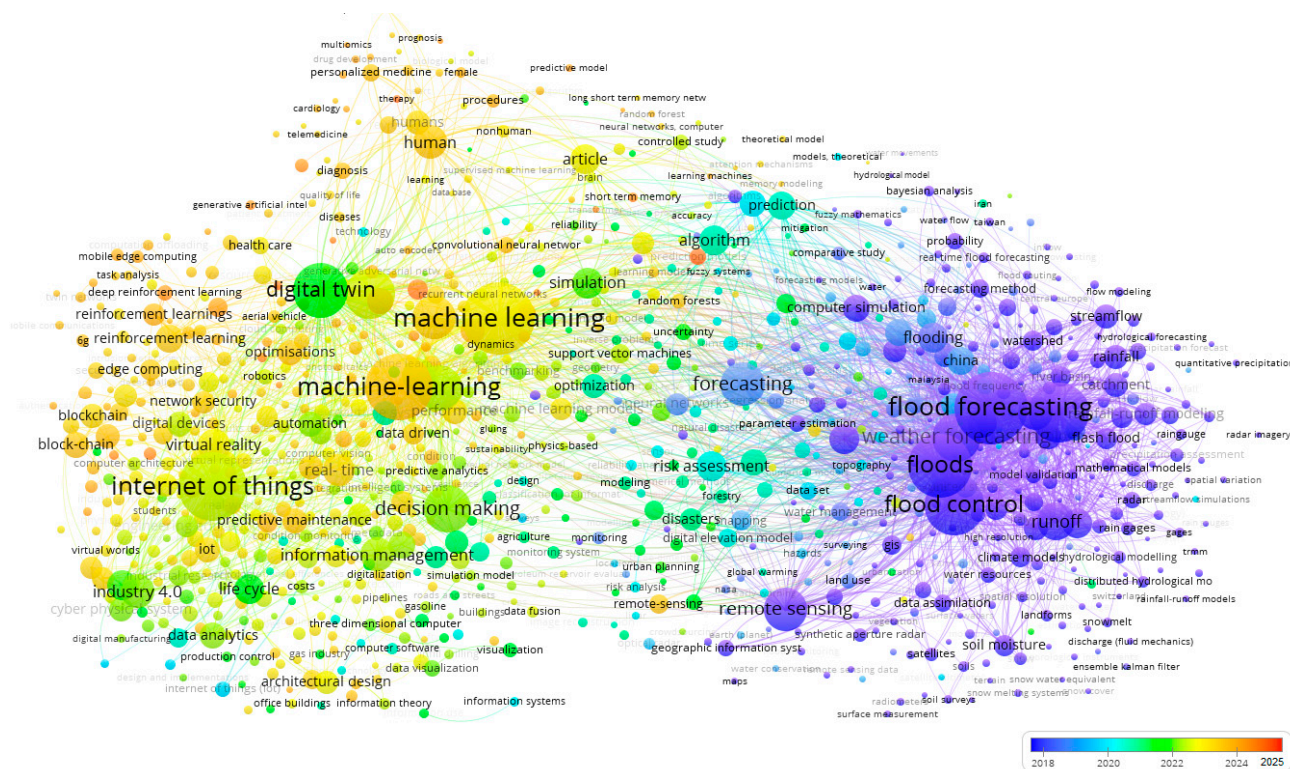


Figure 1. Keyword co-occurrence network for Digital Twin applications in urban flood risk management (2018–2025).

The search strategy employed Boolean logic to combine targeted keywords such as “Digital Twin”, “urban flood risk management”, “remote sensing”, “IoT”, “artificial intelligence”, and “climate extremes” to filter results toward the intersection of digital technologies and hydrological risk mitigation.

Boolean Search Query in Scopus databases and Web of Science conducted in this study: ALL (“Digital Twin” OR “Virtual Replica”) AND (“urban flood risk management” OR “Flood Forecasting” OR “Disaster Resilience”) AND (“Remote Sensing” OR “IoT” OR “AI” OR “Machine Learning”) AND (“Climate Extremes” OR “Hydrological Modelling” OR “Inundation Mapping”))

All search results from Scopus and Web of Science were exported in RIS format and merged in Zotero. Deduplication was conducted in two steps: (i) DOI-based exact match removal, and (ii) fuzzy-title matching (Levenshtein ratio ≥ 0.9) combined with identical first author and publication year. Scopus records found in Web of Science were removed at this stage. The deduplicated set formed the pool for title/abstract screening.

2.3. Inclusion and Exclusion Criteria

The combined database search returned 1085 records (Scopus = 85; Web of Science = 1000). After removing $n = 850$ duplicates, $n = 235$ unique records remained for title/abstract screening. We excluded $n = 150$ based on irrelevance to Digital Twin applications in UFRM, leaving $n = 85$ for full-text review. No records were excluded at the full-text stage. A PRISMA flow diagram (Figure 2) details counts and reasons at each step. The combined database search returned 1085 records (Scopus = 85; Web of Science = 1000). After removing $n = 850$ duplicates, $n = 235$ unique records remained for title/abstract screening. We excluded $n = 150$ based on irrelevance to Digital Twin applications in UFRM, leaving $n = 85$ for full-text review. No records were excluded at the full-text stage. A PRISMA flow diagram (Figure 2) details counts and reasons at each step.

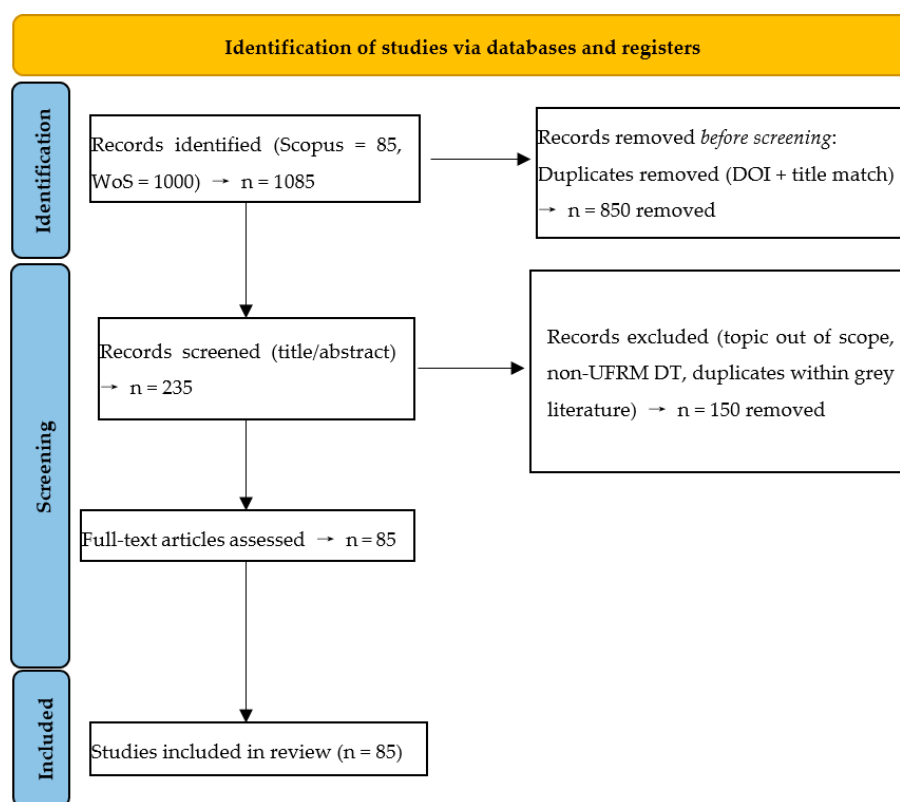


Figure 2. PRISMA 2020 flow diagram for study selection process in Digital Twin applications for urban flood risk management (2018–2025).

Stage 1: Title and Abstract Screening involved the exclusion of studies that lacked direct relevance to the review's hydrological and climate adaptation focus. Specifically, articles were omitted if they addressed peripheral domains (e.g., aerospace engineering, digital healthcare systems), mentioned Digital Twin (DT) concepts without operationalization within urban flood risk contexts, or constituted theoretical expositions devoid of empirical validation. This stage was crucial for refining the corpus toward technically grounded and domain-specific contributions.

Stage 2: Full-Text Evaluation applied a refined set of inclusion criteria prioritizing:

- The articulation of a clear and operational DT framework or architecture,
- Explicit application to urban flood monitoring, early warning systems (EWS), hydraulic modeling, or climate resilience planning,
- Integration of remote sensing (RS) platforms, Internet of Things (IoT) sensor networks, or Artificial Intelligence/Machine Learning (AI/ML) methodologies,
- Engagement with policy frameworks, community participation mechanisms, or equity considerations within disaster risk governance,
- Publication within peer-reviewed outlets with demonstrably transparent methodological sections, enabling reproducibility and validation.

Following rigorous application of these criteria and the removal of duplicates, 85 articles were selected for final inclusion illustrating in Figure 2. This curated dataset not only captures the technological frontier of DT applications in urban hydrology but also situates these advancements within broader discourses of climate governance, urban resilience, and adaptive risk management. The final selection ensures that the systematic review is anchored in a methodologically robust, interdisciplinarity informed, and scientifically impactful body of evidence.

2.4. Data Extraction and Coding

The selected studies underwent a structured, multi-layered analysis aimed at systematically identifying recurring patterns in Digital Twin (DT) applications for Urban Flood Risk Management (UFRM). A mixed-methods framework was employed, combining quantitative frequency analysis and qualitative thematic coding, to extract both statistical trends and nuanced contextual insights.

The core dimensions of the analysis included (Figure 3):

- **Digital Twin Architectures:** Studies were scrutinized for architectural modularity, specifically the separation of DT systems into ingestion, processing, simulation, and decision-support layers, as well as the integration depth across heterogeneous data streams (e.g., remote sensing, IoT, hydraulic models) and the deployment of real-time processing capabilities via edge computing or federated systems.
- **Remote Sensing Technologies:** Diverse RS sources were categorized, encompassing optical satellites (e.g., Sentinel-2, Landsat series, MODIS), UAV-acquired high-resolution imagery, and advanced multispectral or hyperspectral platforms. The technological trade-offs in spatial resolution, revisit frequency, and deployment costs were analyzed, with a critical lens on the disparities between high-income and low-resource settings.
- **AI/ML Model Applications:** The review documented the prevalence of supervised algorithms (e.g., Random Forest, XGBoost), deep learning models (e.g., CNN, U-Net, LSTM networks), and unsupervised techniques while highlighting emergent innovations such as hybrid AI-physics models (e.g., PINNs) for improved flood simulation in data-scarce regions.
- **Application Scales:** DT deployments were mapped across different operational scales, ranging from localized urban applications to watershed-wide models, national infrastructure resilience frameworks, and transboundary river basin management systems.

- Governance and Policy Integration: Special emphasis was placed on the extent of policy linkage whether DTs were embedded within national climate strategies, open data initiatives, or participatory governance models that engage vulnerable communities and prioritize climate justice.

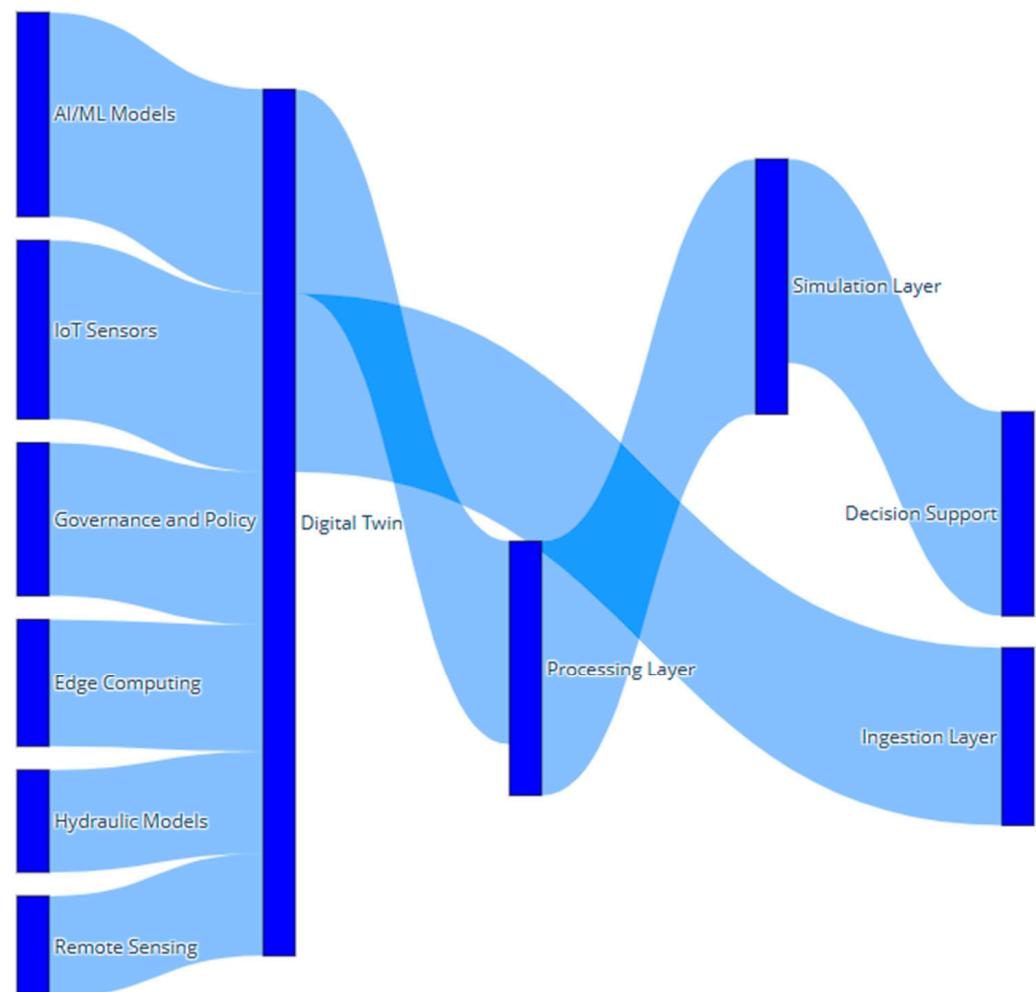


Figure 3. Integration of remote sensing, AI/ML models, and governance frameworks within Digital Twin architectures for climate resilience and decision support.

To ensure a consistent and replicable extraction process, a standardized extraction template was developed, capturing critical metadata such as (Figure 4):

- Authors, publication year, and journal outlet,
- Geographic focus and flood type (pluvial, fluvial, coastal),
- Remote sensing sources and data resolutions,
- AI/ML methodologies applied,
- Degree of stakeholder and community involvement,
- Key technological outcomes and reported challenges.

This rigorous approach enabled both macro-level synthesis (e.g., global geographic trends, technology adoption rates) and micro-level thematic discovery (e.g., novel edge-AI integrations, equity gaps in data access).

By coupling structured data extraction with inductive-deductive thematic coding, the review achieved a high degree of analytical depth, methodological transparency, and scientific reproducibility, setting a strong foundation for evidence-based insights into the evolution of Digital Twin technologies for flood resilience.

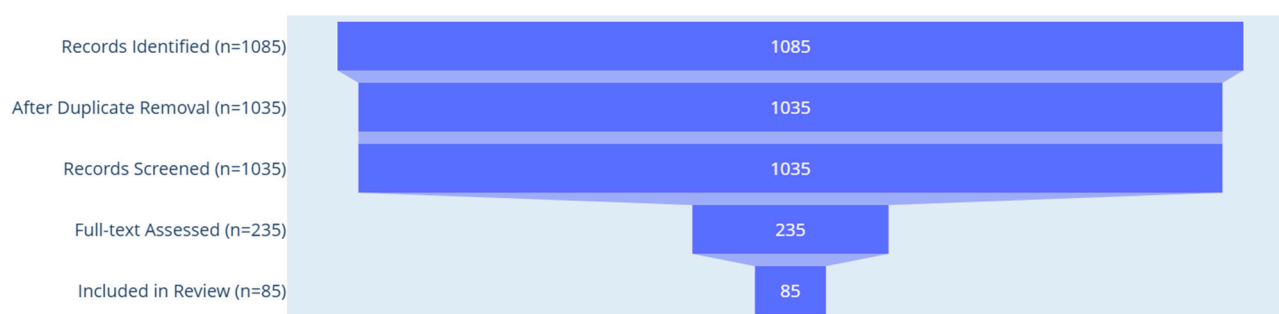


Figure 4. Study selection funnel chart illustrating article counts at each data extraction and coding phase.

2.5. Quality Assessment Criteria

To ensure methodological rigor and scientific impact, a validated five-dimensional quality assessment rubric was developed in accordance with PRISMA 2020 guidelines and best practices for systematic reviews. This rubric evaluates studies across the dimensions outlined in Table 1:

Table 1. Evaluation framework for assessing urban flood risk management (UFRM) solutions: dimensions, scoring criteria, and indicators.

Dimension	Scoring Guide	Indicators
1. Technical Rigor	0 = Weak or absent methods 0.5 = Moderate rigor, some validation 1 = Strong empirical design with validated models	- Use of simulations, experiments, or case studies - Validation of DT models (e.g., against real-time data or historical records) - Computational reproducibility or model benchmarking
2. Innovation	0 = No novel contribution 0.5 = Moderate innovation (e.g., common ML model) 1 = Novel algorithm, architecture, or RS integration	- Introduction of new algorithms (e.g., hybrid AI, physics-informed ML) - Innovative RS-IoT fusion or real-time streaming design - Novel system architecture or deployment setting
3. Relevance to UFRM	0 = Vague or tangential relevance 0.5 = Partial relevance (e.g., smart cities without flood-specific focus) 1 = Direct application to flood prediction, management, or resilience	- Clear flood management use case (forecasting, mapping, EWS) - Urban flood context emphasized - Relevance to hydrological modeling or risk assessment
4. Policy Integration and Equity	0 = No mention of policy or equity 0.5 = Generic mention or theoretical discussion 1 = Concrete discussion of governance, public participation, or social inclusion	- Integration with national or local flood policy - Discussion of community-based DT use - Consideration of vulnerable populations, equity, or climate justice v
5. Transparency and Replicability	0 = No methods or data sharing 0.5 = Partial or unclear sharing 1 = Fully transparent methodology and open tools	- Availability of source code, datasets, or workflow - Clear documentation of tools and processing steps - Use of open-source platforms or public repositories

Each study is assigned a Scientific Robustness Score (SRS) calculated as follows:

$$SRS = \sum_{i=1}^5 D_i$$

where D_i represents the score (0–1) for each dimension. Studies with $SRS \geq 4$ were retained, ensuring alignment with high-impact scientific standards.

Table 2 operationalizes the Scientific Robustness Score (SRS) by illustrating its application to three contrasting studies. This approach clarifies not just how scores are assigned, but how they reflect methodological integrity and translational potential. High-SRS studies, such as Li et al. [33] integrated validated real-time datasets, innovative hybrid AI–physics models, and explicit UFRM applications, creating a blueprint for reproducible and scalable research. By contrast, lower-SRS works revealed structural weaknesses: reliance on closed-source tools, omission of equity considerations, and insufficient validation against independent data. This demonstrates that SRS is not a passive metric but a diagnostic instrument one that can guide authors toward enhancing transparency, replicability, and societal relevance in DT research.

Table 2. Example SRS scoring for selected studies.

Study ID	Technical Rigor	Innovation	Relevance to UFRM	Policy Integration and Equity	Transparency and Replicability	SRS (Total)	Notes
Li et al. [33,34]	1	1	1	0.5	1	4.5	Validated model with real-time data; novel hybrid AI model.
Barrile et al. [35]	0.5	0.5	1	0	0.5	2.5	Moderate rigor; limited equity focus; proprietary software.
Roudbari et al. [36]	1	0.5	1	1	0.5	4.0	Strong policy integration; open datasets; modest innovation.

3. Results

3.1. Five-Dimension Quality Assessment of UFRM Digital Twin Studies

The scientific robustness of the 85 reviewed studies was evaluated using a validated five-dimension rubric encompassing Technical Rigor, Innovation, Relevance to Urban Flood Risk Management (UFRM), Policy Integration and Equity, and Transparency and Replicability [37,38]. Each dimension was scored equally, reflecting balanced importance across methodological and governance considerations [39–41]. The resulting radar chart (Figure 5) highlights strong performance in technical rigor and relevance, with comparatively lower scores in policy integration and transparency, revealing persistent gaps in equity-oriented design and open science practices (Figure 5) [42–44].

3.2. Digital Twin Architectures and Remote Sensing Integration

Digital Twin (DT) architectures applied to Urban Flood Risk Management (UFRM) vary in structural complexity, scalability, and the degree to which they integrate multisource data [45,46]. Most contemporary systems adopt a modular design, typically organized into four core layers: data ingestion, processing, simulation, and decision support [47]. These layers facilitate the real-time integration of diverse spatial and temporal datasets, enabling both predictive and responsive capabilities. Cutter et al. [48] presents a generalized architecture commonly adopted in UFRM-focused Digital Twins (Figure 6).

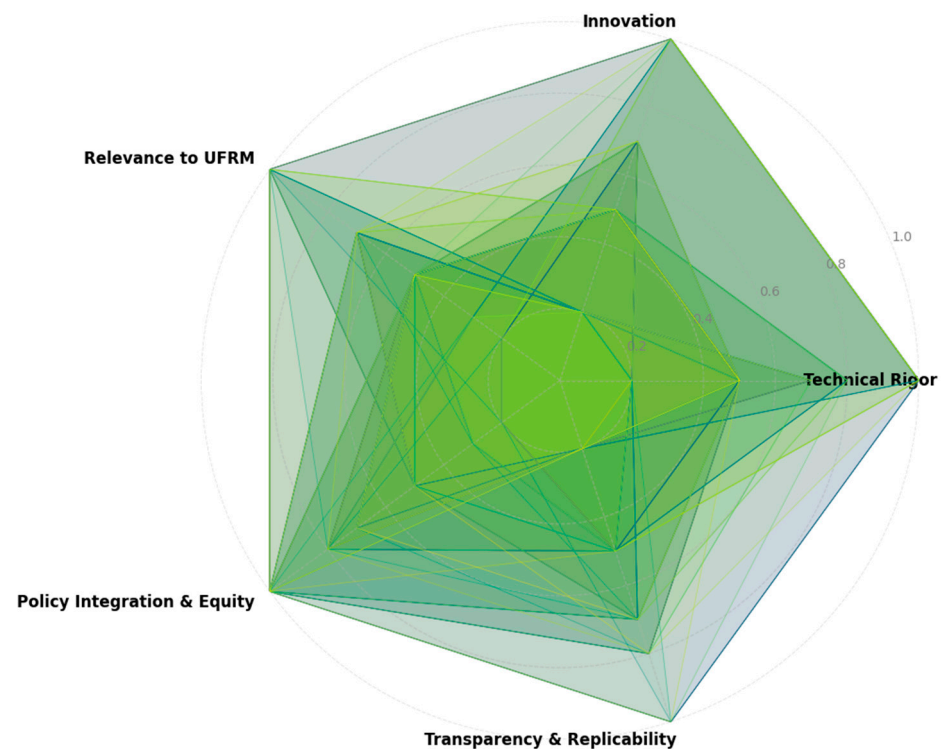


Figure 5. Radar chart of scientific quality assessment of 85 reviewed papers based on five evaluation criteria: technical rigor, innovation, relevance to urban flood risk management (UFRM), policy integration and equity, and transparency and replicability.

At the core of these systems lies the integration of remote sensing (RS) technologies, which serve as the primary input layer for many DT platforms. RS sources are particularly valued for their broad spatial coverage, repeatability, and capacity to support near-real-time monitoring. Key satellite and airborne platforms include:

- Sentinel-1 (SAR): Provides 10–20 m resolution synthetic aperture radar imagery, which is unaffected by cloud cover and critical for flood mapping during extreme weather events. Its revisit frequency and night-time imaging capabilities make it indispensable for emergency response.
- Sentinel-2 and Landsat-8/9: Facilitate the monitoring of vegetation dynamics, soil moisture variability, and impervious surface extent. These variables are crucial inputs for estimating flood susceptibility indices.
- Unmanned Aerial Vehicles (UAVs): Offer sub-meter spatial resolution (0.1–0.5 m), making them ideal for fine-scale assessments in densely urbanized floodplains. UAV-based mapping is particularly valuable for post-flood damage assessment and calibration of hydrodynamic models.
- LiDAR (Light Detection and Ranging): Delivers highly accurate digital elevation models (DEMs) with cm-level vertical precision. Urban DTs such as that implemented in New York City leverage LiDAR to capture terrain features, drainage networks, and microtopography essential for simulating overland flow paths.

The integration of Internet of Things (IoT) technologies further enhances temporal granularity and local-scale situational awareness. Smart sensors, deployed in storm drains, rivers, and roadways, feed live data streams into DT platforms. A notable example comes from Kenya, where a low-cost IoT sensor network built on Raspberry Pi microcontrollers led to a 40% reduction in flood warning times, significantly improving disaster preparedness in informal settlements.

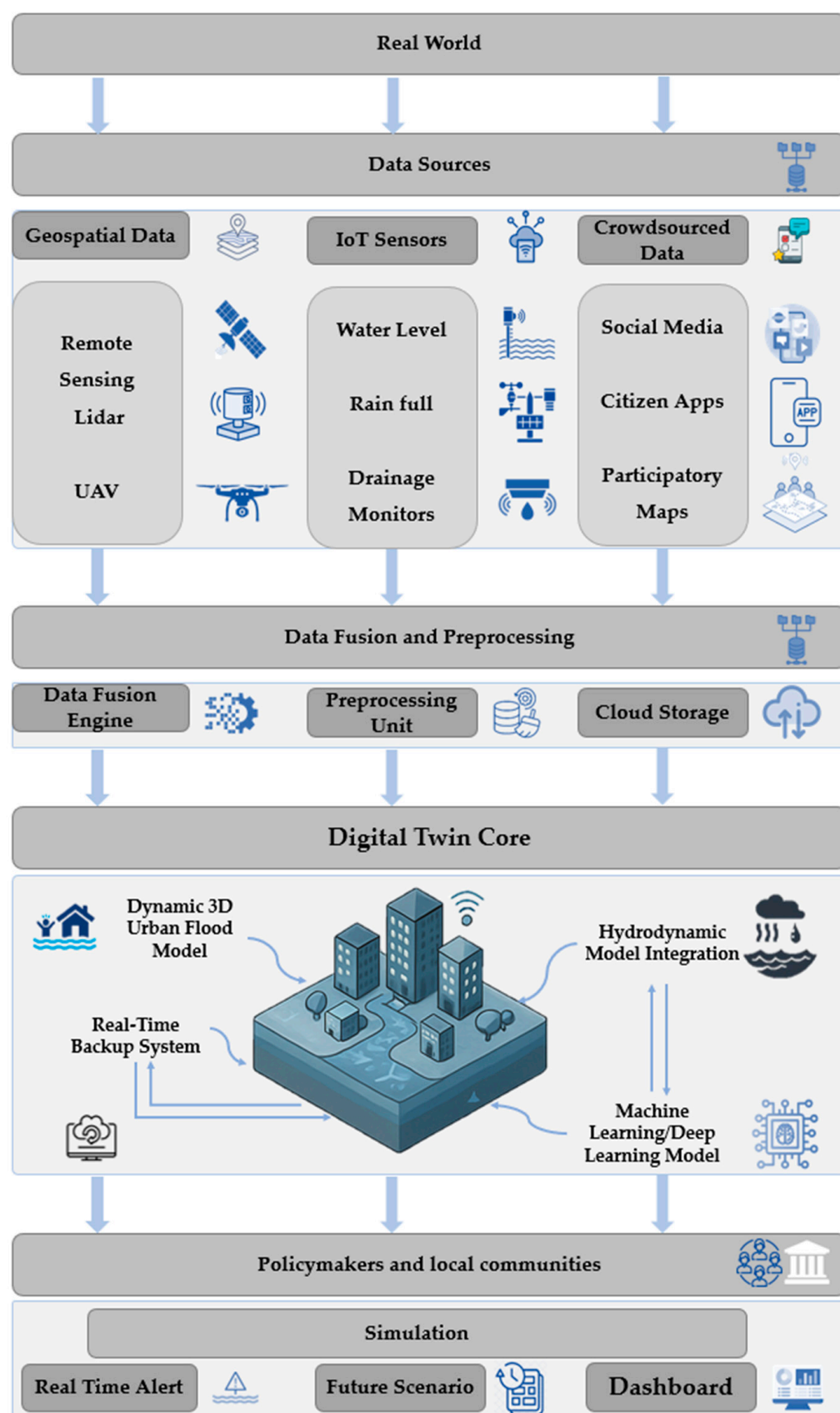


Figure 6. Architecture of Digital Twins for urban flood risk management.

Advanced machine learning (ML) models augment these systems by enabling real-time classification, predictive analytics, and anomaly detection. ML models support both

pre-event risk assessment and dynamic flood forecasting. Tables 3 and 4 summarize widely used ML/AI approaches in flood-oriented DTs, alongside their respective application domains and notable studies.

Table 3. Common AI/ML models used in flood-related Digital Twin systems.

Model Type	Application Area	Notes	Example Study
Transformer-based Models	Compound hazard forecasting	Useful for multi-input time series (e.g., cyclone + flood)	[49]
Physics-Informed Neural Nets	Hydraulic modeling	Enforce boundary constraints in simulations	[50]
Random Forest, XGBoost	Land cover change, flood susceptibility	Strong performance with RS datasets	[51]
U-Net and CNN variants	Flood extent mapping from SAR	High accuracy in image segmentation tasks	[52]

Table 4. Comparative effectiveness of DT technology configurations for different flood types.

Flood Type	Best Sensors	Best AI/ML Models	Simulation Resolution	Strengths	Weaknesses
Urban Flash Flood	UAV, Sentinel-1 SAR	U-Net, CNN	<1 m	High spatial precision; rapid deployment	High UAV cost; limited coverage
Riverine	Landsat, Sentinel-2, LiDAR	RF, XGBoost, PINNs	5–10 m DEM	Strong for large catchments; good temporal coverage	Coarser urban detail
Coastal	Sentinel-1 SAR, altimetry	LSTM, Transformers	10–30 m DEM	Captures tidal/surge events	Lower inland accuracy

Comparative assessment [53] highlights distinct optimization pathways: urban flash floods benefit most from sub-meter UAV or SAR imagery coupled with deep learning (U-Net, CNN), while riverine inundation favors multispectral data with physically informed ensemble models. Coastal flooding demands altimetry and temporal sequence models (LSTM, Transformers) to capture surge dynamics. These patterns suggest that “one-size-fits-all” DT configurations are suboptimal; instead, hazard-specific architectures should be prioritized in system design.

To demonstrate a real-world integration pathway, Figure 7 illustrates a data flow schematic of a remote-sensing-integrated DT used in Calgary, Canada [54]. The system combines Sentinel-1 SAR imagery, live sensor data, and city-level hydrodynamic models to simulate inundation and generate decision support outputs for emergency response teams [55].

The effectiveness of these architectures depends on robust data fusion techniques and real-time data assimilation methods [56]. As such, ongoing developments in deep learning, edge computing, and spatiotemporal modeling are poised to play a central role in the next generation of flood-aware Digital Twins.

3.3. Global Applications and Case Studies

Digital Twin (DT) technologies for Urban Flood Risk Management (UFRM) have seen rapid proliferation across global regions, each shaped by unique environmental, socio-political, and technological conditions. These implementations vary significantly in terms of data resolution, modeling sophistication, computational capacity, and integration with governance frameworks. Nevertheless, they collectively demonstrate the versatility of DTs

as a unifying framework for anticipatory flood modeling, real-time decision-making, and resilient urban planning [57,58].

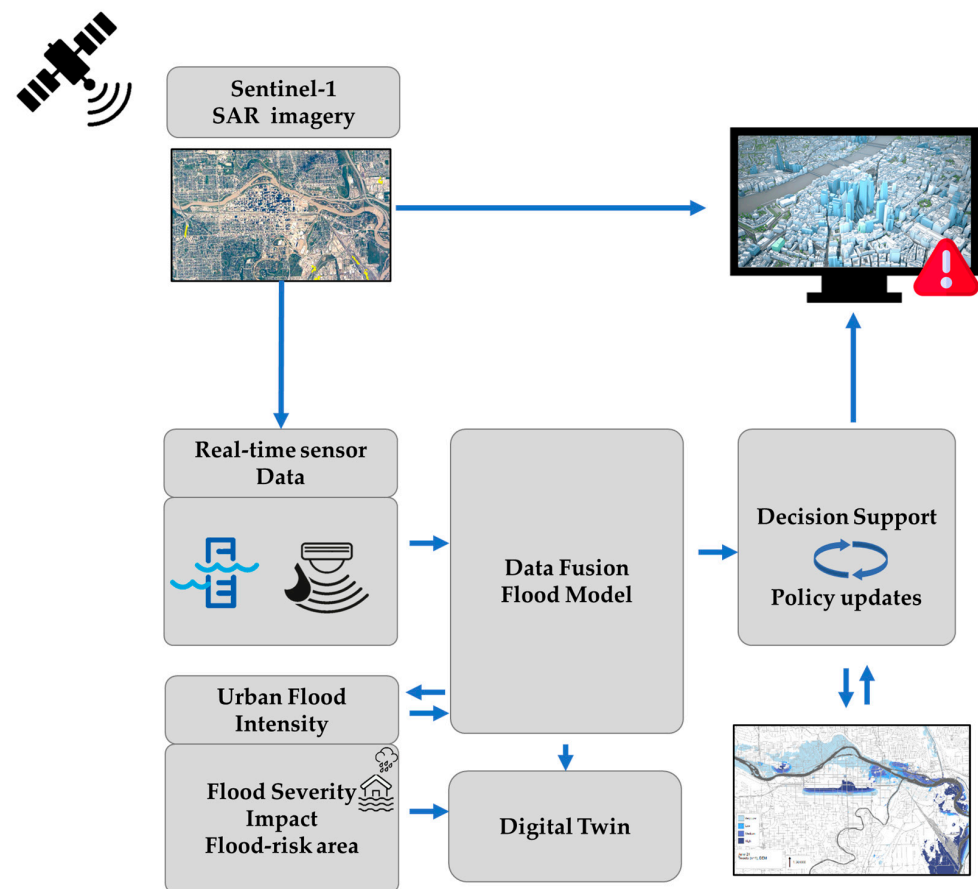


Figure 7. Data flow of a remote-sensing-integrated DT for UFRM.

The regional distribution (Figure 8) confirms a heavy geographic skew, with 57% of studies originating from Europe and high-income Asia, and only 26% from LMIC regions. This imbalance is statistically significant given the disproportionate flood burden in LMICs, reinforcing prior claims of a digital divide. The scarcity of African, Latin American, and LMIC Asian implementations limits the external validity of DT frameworks designed in high-resource contexts [59].

Figure 9 provides a comparative synthesis of key Digital Twin applications for UFRM, highlighting the diversity in platforms, technological stacks, and achieved impacts.

These case studies underscore the role of Digital Twins in transforming flood risk governance from reactive to predictive paradigms [60,61]. In Italy, integration of SAR data, GNSS and cloud computing facilitated rapid economic loss estimation during the mega-floods, enabling more targeted and efficient humanitarian interventions [62,63]. Similarly, Japan's River Basin Digital Twins, developed as part of its water security agenda, exemplify large-scale, multi-basin integration enabled by high-performance computing and sensor fusion [64,65].

Table 5 separates systematically coded results from narrative case study detail, revealing technology convergence patterns across regions. Urban implementations predominantly employed high-resolution RS (SAR, UAV) with AI-based image segmentation, while large river basin models favored multispectral data and ensemble ML methods. The separation of results from narratives ensures analytical clarity and enables practitioners to compare configurations without the confounding influence of context-heavy storytelling.

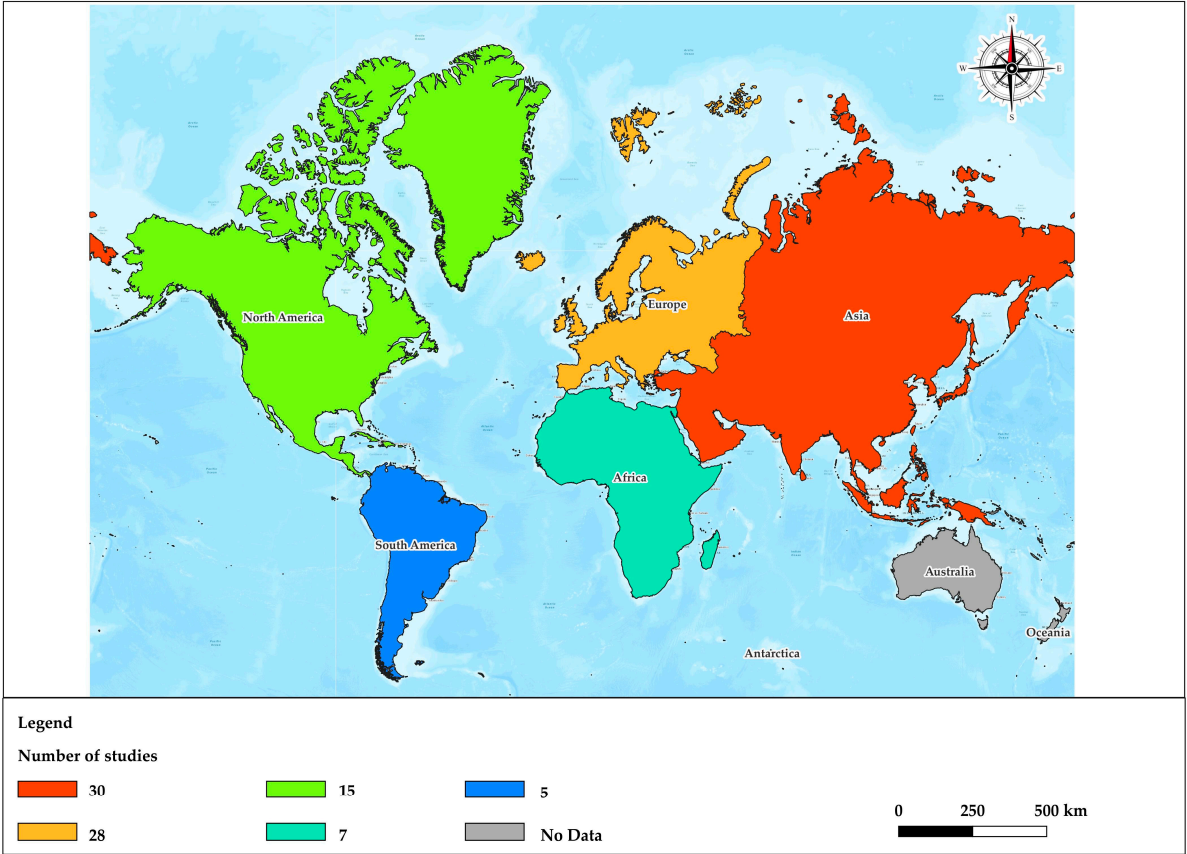


Figure 8. Regional distribution of reviewed studies.

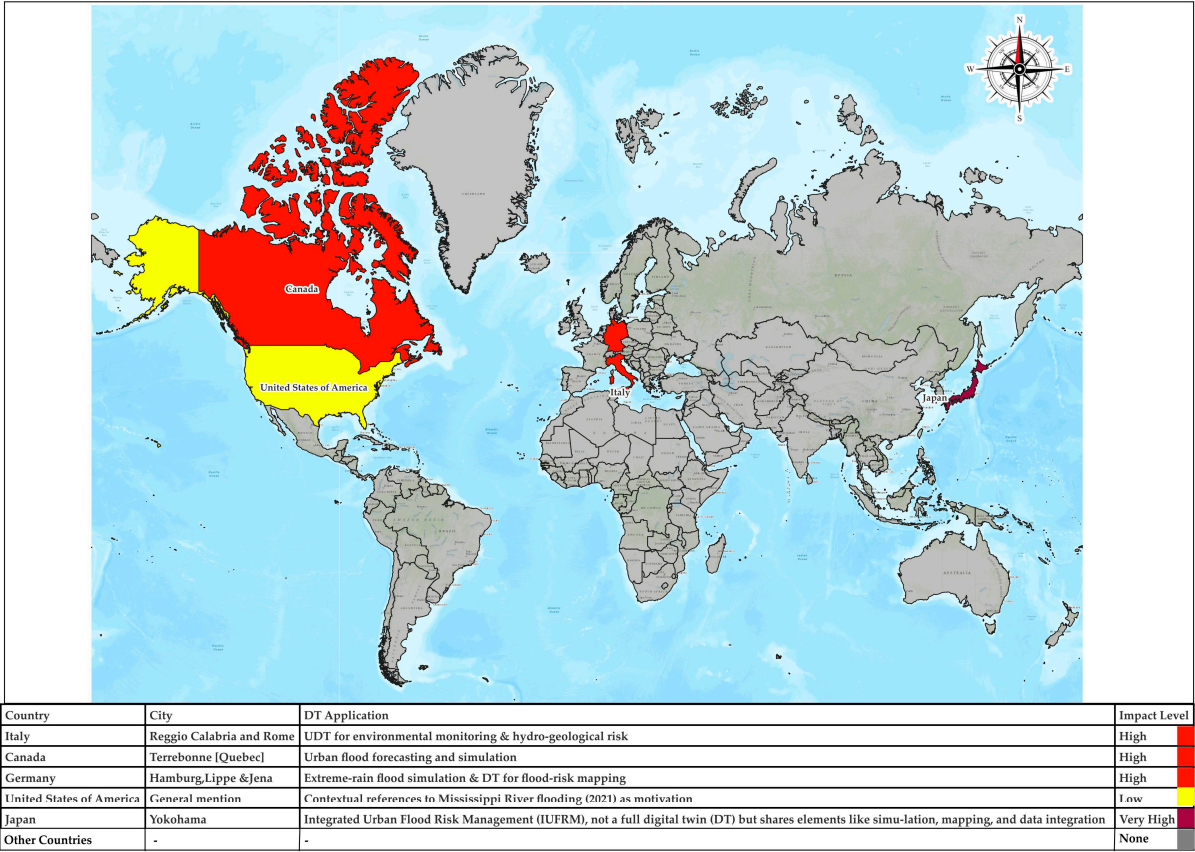


Figure 9. Summary of global Digital Twin implementations in urban flood risk management.

Table 5. Summary of DT implementations in reviewed studies.

Country/City	Technologies Used	Flood Type	Reported Impacts
Yokohama, Japan	GIS, hydrological modeling, hazard mapping	Pluvial and coastal	Multi-layered mitigation strategies; real-time support.
Hamburg, Germany	DT, hydrodynamic models, infrastructure data	Pluvial	Improved extreme rain event forecasting.
Calgary, Canada	DT, SAR, IoT, hydrodynamic simulation	Riverine	Accurate water level forecasts 9 days ahead.

Europe’s advanced implementations, such as Germany’s hydrodynamic DTs for dam failure forecasting, reflect the region’s long-standing investment in flood modeling and real-time simulation [66,67]. These systems demonstrated tangible impacts by reducing warning lead times and improving public evacuation strategies during heavy floods [68–70].

In lower-resource contexts like developing countries, innovative fusions of Sentinel-2 satellite imagery with community-driven knowledge systems have created hybrid DTs that function despite sparse gauging networks [71–73]. Such systems are critical for increasing equity in flood preparedness in the Global South (Developing countries) [74].

The New York City DT initiative represents a frontier application of high-resolution LiDAR, crowdsourced data (e.g., Twitter), and 3D GIS platforms for asset-level flood vulnerability modeling [75,76]. Its outputs inform both emergency response and long-term resilience planning (Table 6).

Table 6. Global distribution of Digital Twin flood implementations (2018–2025).

Country	City	Implementation Density	Notes	Source
Portugal	Lisbon	High	Advanced adoption; integrates IoT, AI, and climate models into flood forecasting systems.	[77]
Finland	Helsinki	High	Pioneers in urban flood resilience using Digital Twins; strong policy support.	[78]
Japan	Tokyo	High	Rapid urbanization driving DT use; smart cities initiatives support flood management systems.	[79,80]
USA	Waterloo	High	Focused on flood risks and resilience in urban planning and emerging interest; limited by infrastructure and funding.	[81]
USA	Ohio	High	The project utilized open-source software and data, incorporating several key components: the generation of flood alerts and detection of flood events; rapid mapping of flood extents and continuous monitoring during active flooding; short-term forecasting of flood coverage and surface elevation using high-resolution hydrodynamic models in targeted areas; and the real-time and post-event evaluation of financial risks associated with flooding.	[82]
Rwanda	-	Emerging	GeoHaTACC is an operational toolbox designed to detect and inventory hydro-geological hazards in tropical regions, while also documenting the impacts of climate change on these hazards. The system integrates multiple data sources and is currently being piloted in Rwanda, a country significantly affected by such events, with the long-term goal of expanding its application to other territories.	[83,84]
Madagascar and Reunion Island	-	Low	The Cimopolée Project focuses on monitoring the impacts of tropical cyclones in the southwest Indian Ocean using satellite Earth observation data. It supports rapid damage assessment and resilience analysis for emergency response and planning. The project contributes to Digital Twin applications by integrating real-time geospatial data into risk management tools.	[85,86]

Incorporating limitations (Table 7) tempers overly optimistic narratives of DT performance. Even high-profile systems like H2Porto and Calgary’s DT face operational constraints ranging from latency during peak events to long-term model drift and public uptake challenges. The recurrence of interoperability issues across projects suggests that technological sophistication alone does not guarantee sustained impact without institutional integration and maintenance planning.

Table 7. Selected DT implementations: Achievements and limitations.

Case Study	Reported Achievements	Limitations/Challenges
H2Porto (Portugal)	Reduced response time by 35%; integrated IoT and RS	Data latency during extreme events; maintenance costs
Calgary (Canada)	Accurate 9-day forecasts; interactive simulations	Model drift; high compute requirements
Reggio Calabria (Italy)	High-precision 3D model; hazard mapping	Low public uptake; interoperability issues

Despite promising advancements, several barriers limit the widespread adoption and effectiveness of DTs in UFRM.

3.4. Data and Infrastructure Inequity

A major barrier to the broad deployment of Digital Twins (DTs) in flood risk management lies in the unequal distribution of data infrastructure and technical resources:

- Currently, less than 12% of African river basins are monitored by operational real-time hydrological stations, limiting the ability to accurately calibrate and validate DT-based models [87,88].
- Although high-resolution remote sensing (RS) data is becoming increasingly accessible, its use in many Global South regions remains minimal due to constraints such as limited internet bandwidth, high cloud service costs, and inadequate data processing expertise [89,90].
- Furthermore, cloud-native platforms critical for enabling real-time simulation, data fusion, and alert generation are often unaffordable or inaccessible due to economic or geopolitical restrictions [91].

These disparities not only reduce the operational effectiveness of DT systems in vulnerable areas but also diminish their trustworthiness among local stakeholders, reinforcing a global digital divide in disaster forecasting, mitigation, and adaptation capabilities.

3.5. Interoperability and Standards

Digital Twin ecosystems are inherently complex, involving the integration of diverse data streams satellite imagery, IoT sensors, hydraulic and hydrodynamic models, and even social media inputs [92]. Yet significant technical fragmentation persists:

- The lack of standardized data formats, shared ontologies, and open application programming interfaces (APIs) impedes seamless integration across platforms and agencies [93,94].
- Discrepancies in spatial and temporal resolution for example, between SAR and optical RS data or between sensor networks and modeling outputs complicate the synchronization of inputs across disciplines [95].
- Moreover, in many national contexts, digital systems are siloed across governmental and municipal institutions, leading to isolated datasets and incompatible workflows that obstruct comprehensive DT development [96].

The lack of technical interoperability not only slows development but also undermines real-time coordination during emergencies [97].

3.6. Computational Burden

DT systems, particularly those simulating fine-grained urban hydrodynamics, require substantial computational power posing a major constraint for many jurisdictions:

- Simulations operating at building-level spatial resolution or over complex terrain often rely on access to high-performance computing (HPC) environments or cloud-based GPU clusters to deliver timely outputs [98].
- Where access to cloud computing or local HPC infrastructure is limited, data-to-decision loops become sluggish or unreliable, significantly compromising their operational relevance for early warning or emergency response [99].
- Although edge computing offers promise for decentralized data processing, its application in UFRM remains in the early stages and is largely untested in rural or low-resource settings [100].

These constraints underscore the need for more energy-efficient algorithms, modular DT architectures, and shared compute infrastructure for climate-vulnerable regions [101,102].

3.7. Policy Fragmentation

Even when Digital Twin technologies are technically viable, institutional and policy-related hurdles can significantly delay or prevent their implementation (Figure 10):

- In cross-border watersheds like the Nile or Mekong, fragmented legal mandates and weak data-sharing frameworks obstruct the formation of integrated DT systems.
- The absence of enabling regulations, limited government incentives for digital innovation, and insufficient funding pathways further discourage institutional investment in DT platforms for flood management.
- In many urban settings, overlapping jurisdictions and decentralized flood management responsibilities produce organizational silos, resulting in low data interoperability and resistance to system-wide reform.

Moreover, the lack of robust legal frameworks governing the use of participatory or crowdsourced data like social media raises ethical and privacy concerns, further fueling institutional reluctance to adopt Digital Twin strategies illustrating in Figure 10.

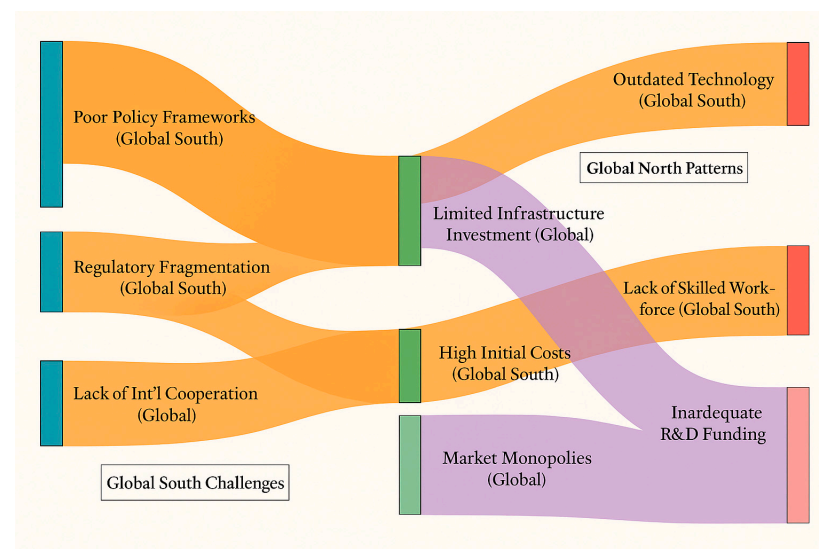


Figure 10. Barriers to Digital Twin scalability in flood risk systems, categorized by type (technical, economic, governance).

4. Discussion

4.1. Comparative Advantages over Traditional UFRM

This review synthesizes emerging trends, technologies, and global applications of Digital Twin (DT) systems in urban flood risk management (UFRM). An analysis of 85 peer-reviewed articles published between 2018 and 2025 highlights the transformative potential of DTs when integrated with remote sensing (RS), the Internet of Things (IoT), and artificial intelligence (AI). Despite these advancements, persistent challenges remain particularly in implementation, equity, and standardization which must be addressed to fully harness DTs for climate resilience.

As a paradigm-shifting innovation, DTs offer real-time simulation, adaptive decision-making support, and integrative modeling across diverse data modalities [103]. The global case studies discussed in Section 3.2 demonstrate the adaptability of DT architectures, ranging from infrastructure resilience planning in New York City to compound hazard forecasting in East Asia. These examples underscore the flexibility of DTs across varying geographic, institutional, and hydrological contexts.

Moreover, in parallel with advancements in data fusion and real-time simulation, a growing body of work explores the convergence of 3D Virtual Reality (VR) and real-time Virtual Reality within Digital Twin systems [104]. These hybrid approaches transcend traditional geospatial visualization by creating immersive, dynamic environments that model real-world urban and natural systems in both structure and function. Real-time VR-enhanced DTs incorporate live data feeds such as sensor outputs, satellite-derived inundation models, and crowd-sourced observations to continuously update a photorealistic, interactive replica of the urban landscape [105]. This integration enables decision-makers, emergency responders, and stakeholders to engage with simulated flooding scenarios in real time, improving public engagement, stakeholder understanding, and inclusive disaster planning. Given the significant data and infrastructure gaps in African countries, and the fact that several co-authors are based in Africa, this review underscores the urgent need for targeted DT research and pilot implementations in African urban centers to address both technological and governance deficits.

4.2. Interoperability and System Design

Despite the promise of DTs, several technical challenges hinder their full potential. The absence of interoperability frameworks and common standards undermines the integrative capacity of DTs [106]. Current deployments remain fragmented due to inconsistent data schemas, incompatible spatio-temporal resolutions, and a lack of shared ontologies or APIs [107]. This fragmentation impairs synchronization of real-time data streams across satellite, sensor, and hydrodynamic domains, compromising system coherence particularly in multi-agency or transboundary contexts [108].

Furthermore, the computational burden associated with DT operations introduces a performance-access paradox [109]. As system fidelity improves, operational accessibility declines. Urban-scale flood simulations that integrate fine-grained topographic data and real-time assimilation demand high-performance computing (HPC) or cloud-native GPU environments [110]. In low-resource settings lacking such infrastructure, the closed-loop capacity of DTs transforming data into action in real time remains more aspirational than actionable [111].

4.3. Geographic and Socio-Technical Inequities

The trajectory of DT diffusion reveals significant structural asymmetries. The uneven global distribution of Digital Twin capacity is not merely a reflection of technological lag

but symptomatic of deeper systemic constraints intersecting technical, institutional, and socio-political dimensions [112].

The design-choice mapping (Table 8) explicitly links flood hazard type to preferred sensor suites, AI models, and simulation resolution. This matrix offers a prescriptive tool for practitioners to align DT system design with hazard characteristics, improving both efficiency and predictive accuracy [113]. Such explicit mappings can also support standardization efforts by embedding hazard-specific configuration baselines into interoperability guidelines [114].

Table 8. Mapping DT design choices to flood types.

Flood Type	Preferred Sensors	AI Models	Simulation Resolution
Urban Flash Flood	UAV, SAR, IoT rain gauges	CNN, U-Net	<1 m
Riverine Flood	Optical RS, SAR, LiDAR	RF, XGBoost, PINNs	5–10 m
Coastal Flood	SAR, tide gauges, altimetry	LSTM, Transformers	10–30 m

First, data and infrastructure inequity represent a foundational barrier. The underrepresentation of real-time hydrological monitoring networks in the Global South (Developing countries), limited access to high-resolution remote sensing data, and costly cloud infrastructure severely constrain the operationalization of DTs in climate-vulnerable regions [115]. These limitations impede model calibration, erode the credibility of DT outputs, and deepen the digital divide in anticipatory disaster governance [116].

Second, governance fragmentation and institutional inertia pose non-trivial barriers to sustained DT adoption [117,118]. In many jurisdictions, flood risk management responsibilities are distributed across siloed agencies, inhibiting data sharing and systemic integration [119]. The lack of enabling policy frameworks particularly those governing data ethics, privacy, and algorithmic accountability further constrains the scalability of DTs, even in technically capable contexts [120].

Together, these structural issues underscore that the future of Digital Twins in UFRM will be shaped less by technological breakthroughs and more by the resolution of systemic constraints [121]. Advancing DT adoption equitably will require a shift from technology-first paradigms to ecosystem-level approaches emphasizing interoperability, institutional reform, capacity building, and equitable access to data infrastructure [122].

4.4. Emerging Ethical Considerations

While immersive Digital Twin systems enable scenario testing and enhanced decision-making, they also raise new ethical and governance challenges [123]. The integration of 3D VR and real-time data amplifies concerns regarding interoperability, latency, bandwidth, and especially ethical governance [124,125].

Key concerns include data privacy, user consent, and the responsible use of immersive media in public risk scenarios [126]. The potential for misrepresentation, misuse of sensitive geospatial data, and unequal access to these tools must be critically examined [127]. As DTs become more immersive and interactive, future research must prioritize the development of open, modular, and scalable platforms capable of supporting both real-time interactivity and ethical integrity especially in data-scarce or infrastructure-poor settings [128,129].

Ultimately, Digital Twins should be reframed not just as technical tools but as sociotechnical infrastructures embedded in the political economies of risk, knowledge, and governance [130]. Co-design with local communities, development of lightweight DT architectures, and establishment of transnational ethical data frameworks will be essential for realizing their transformative potential in urban flood governance [131,132].

4.5. Critical Reflections and Review Limitations

Despite this, this systematic review is based on methodological rigor; some structural limitations and knowledge asymmetries demand a close inspection. These constraints not only define the limits of the present evidence base but also highlight systematic gaps that future research has to rapidly cover. First, the limitation to English-language peer-reviewed literature might have unintentionally omitted important studies carried out and released in other main world languages. Particularly marginalizing important insights emerging from non-Anglophone areas with acute flood vulnerability and active Digital Twin (DT) innovation such as China, Latin America, and parts of West and Central Africa this linguistic filter imposes a kind of epistemic bias [133]. Correcting this error requires intentional inclusion in future reviews of multilingual databases and translation-supported synthesis techniques. Many national DT systems, particularly those linked to important infrastructure or sensitive security areas, limit access to complete technical details, validation methods, and governance systems because they are proprietary or classified [134]. Lack of openness makes peer benchmarking difficult as well as the assessment of scalability, ethical protections, and inclusiveness [135]. Without access to such material, the global DT debate runs the danger of being shaped by a distorted sample of freely available, mostly Western case studies [136]. Third, temporal constraints still apply. Although Scopus and Web of Science offer thorough scholarly coverage, their indexing lag means that innovative ideas, especially in generative artificial intelligence, real-time edge computing, and federated modeling may be under-represented. Review methods need to adapt to include up-to-date databases and non-traditional sources of knowledge as the pace of innovation at DT-UFRM increases while still maintaining scientific quality [137]. The literature's spatial and equity aspects expose clear asymmetries. Though flood risk is a global concern, most empirical research is focused in wealthy, data-rich environments. Indexed research still mostly hides grassroots innovations, indigenous knowledge systems, and community-driven DT deployments in the Global South (Developing countries) [138]. Along with distorting the evidence terrain, this erasure compromises the normative aim of climate justice ingrained in agendas of global disaster resilience [139]. Understanding these constraints is not a weakness but rather a strategic need. They directly guide the priorities stated in the following section and expose important blind areas in present knowledge systems. By facing these gaps, future research can only develop truly inclusive, accessible, and effective Digital Twin frameworks for flood resilience in an era of growing climate crises [140].

4.6. Future Research Directions

In light of the critical reflections presented, future research on Digital Twin (DT) applications must advance beyond theoretical abstraction and narrow technical optimization [141]. A persistent limitation across the reviewed literature is the overemphasis on conceptual frameworks, simulation capabilities, and algorithmic novelty often at the expense of real-world validation, deployment feasibility, and stakeholder integration [142]. Many DT studies remain confined to controlled experimental settings or proof-of-concept architectures, lacking empirical linkage to functioning urban systems, governance workflows, or decision-making processes under uncertainty [143].

To bridge this theory–practice divide, research must prioritize translational scalability transforming sophisticated models into operational tools for city planners, emergency managers, and local communities [144]. This necessitates co-designed pilot implementations with municipal authorities, iterative testing during actual flood events, and performance benchmarking across diverse infrastructural, hydrological, and socioeconomic conditions [145].

Equally vital is the expansion of the epistemic and geographic scope of DT research. This includes fostering multilingual, cross-cultural, and participatory methodologies that amplify the voices and priorities of underrepresented regions particularly those in the Global South (Developing countries) facing disproportionate flood exposure yet underrepresented in high-impact DT literature [146]. Collaborative frameworks should integrate indigenous knowledge systems, local hazard narratives, and community-sourced data to inform more contextually grounded and just modeling approaches [147].

On the technical front, future DT systems must be modular, open-access, and computationally frugal [148]. Architectures that are deployable on edge devices, resilient to intermittent connectivity, and designed with minimal infrastructural requirements will be crucial for low-resource environments [149]. Moreover, interoperability protocols covering ontologies, application programming interfaces (APIs), and data schemas must be standardized to enable seamless integration with municipal information systems and early warning platforms [150].

From a policy perspective, Digital Twins must evolve into decision-support ecosystems. This entails embedding interactive dashboards, scenario planning modules, and equity metrics into DT platforms to enhance usability by non-technical actors and to align outputs with real-time governance needs [151]. Transparency, explainability of AI components, and inclusive co-design processes will be key to institutional trust, adoption, and long-term urban resilience planning [152].

The Technology Readiness Level (TRL) assessment (Table 9) provides a pragmatic filter for emerging DT concepts [153–155]. Crowdsourced geospatial data scores highest (TRL 6) in operational readiness, given low infrastructure demands and existing mobile penetration in LMICs [156,157]. Conversely, VR-enhanced DTs remain mid-range (TRL 5) due to bandwidth and hardware constraints, while hybrid AI-physics models, despite high potential accuracy, remain at early integration stages (TRL 4) due to computational overhead [158].

Table 9. Emerging DT trends and readiness.

Trend	Current TRL (1–9)	Operational Constraints	Potential in LMICs
VR-enhanced DTs	5	High bandwidth; costly hardware	Low–medium
Crowdsourced geospatial data	6	Data validation; uneven participation	High
Hybrid AI-Physics models	4	High computational demand	Medium

Moreover, advancing DTs for urban flood risk management requires a paradigm shift: from building more technologically complex models to building more usable, scalable, and socially responsive systems anchored not only in innovation, but also in impact, equity, and long-term sustainability.

5. Conclusions

This review presents a comprehensive synthesis of the current state and global evolution of Digital Twin (DT) technologies in Urban Flood Risk Management (UFRM), drawing from 85 peer-reviewed publications between 2018 and 2025. The analysis illustrates that DTs are redefining the landscape of flood risk governance by enabling proactive, real-time, and integrated responses through the convergence of remote sensing (RS), Internet of Things (IoT), and artificial intelligence (AI).

DT systems offer considerable advantages over traditional flood management approaches. They enhance the precision of flood forecasting, facilitate faster emergency response through early warning capabilities, and support dynamic scenario planning

for infrastructure and policy decision-making. Case studies from advanced urban center provide compelling evidence of DT effectiveness, particularly when immersive visualization tools and 3D platforms are deployed to engage stakeholders and support interactive, informed governance.

Despite their promise, DT deployment faces several systemic barriers. Disparities in data access and technical infrastructure remain stark, especially in low-resource regions where real-time monitoring networks and high-resolution satellite data are limited. The absence of common technical standards and interoperability frameworks makes it difficult to integrate diverse data sources across platforms and institutions. Additionally, the computational intensity of DT systems restricts their real-time functionality in settings without access to high-performance computing resources.

Institutional challenges also hinder widespread adoption. Many jurisdictions exhibit fragmented governance structures and regulatory uncertainty, leading to siloed decision-making and resistance to digital innovation. As DTs become increasingly immersive and AI-powered, ethical concerns around data privacy, algorithmic bias, and equitable access are gaining urgency but remain insufficiently addressed in most implementations.

To harness the full potential of DTs in UFRM, future efforts must focus on developing modular and open-source frameworks that are accessible and adaptable to low-resource environments. Data integration standards must be established to enable seamless collaboration across jurisdictions, and participatory design practices should be employed to include local knowledge and community priorities. Equally important is the alignment of DTs with governance systems to ensure institutional capacity, policy coherence, and long-term sustainability. Ethical principles must underpin the development and deployment of DTs to promote transparency, accountability, and inclusivity.

Ultimately, Digital Twins should be understood not merely as high-tech modeling tools, but as complex sociotechnical systems embedded in the political, institutional, and ethical dimensions of climate governance. Their ability to contribute meaningfully to urban flood resilience depends on more than technological sophistication—it requires a shift toward equity-centered, context-aware, and collaboratively governed solutions. If pursued with these values at the core, DTs offer a transformative path toward anticipatory, just, and sustainable urban futures in an era marked by intensifying climate risk.

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Abbreviations

The following abbreviations are used in this manuscript:

DT	Digital Twin
UFRM	Urban Flood Risk Management
RS	Remote Sensing
IoT	Internet of Things
AI	Artificial Intelligence
ML	Machine Learning

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